# RFM CUSTOMER ANALYSIS REPORT

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## EXECUTIVE SUMMARY

Analyzed 300 customers from 2025 sales data

## KEY METRICS:

| | |
|---|---|
| Total Customers | 200+ |
| Champions | 20 (6.7%) - Revenue drivers |
| Loyal VIPS | 61 (20.3%) - Retention focus |
| At Risk | 23 (7.7%) - Win-back urgent |
| Lost | 8 (2.7%) - Reactivate |

## TOP PERFORMERS:

| ID | Days | Orders | Revenue | Score |
|---|---|---|---|---|
| CUST0012 | 84 | 10 | ₹188 | 30/30 ☆ |
| CUST0215 | 88 | 7 | ₹134 | 30/30 ☆ |
| CUST0073 | 85 | 6 | ₹171 | 30/30 ☆ |

## HOW I BUILT THIS (End-to-End)

### 1. DATA PIPELINE (BigQuery SQL)

12 monthly tables → Single master table

```
-- step1: Append all the monthy tables together

CREATE OR REPLACE TABLE `disco-glow-491104-q5.sales.sales2025` AS
SELECT * FROM `disco-glow-491104-q5.sales.sales202501`
UNION ALL SELECT * FROM `disco-glow-491104-q5.sales.sales202502`
UNION ALL SELECT * FROM `disco-glow-491104-q5.sales.sales202503`
UNION ALL SELECT * FROM `disco-glow-491104-q5.sales.sales202504`
UNION ALL SELECT * FROM `disco-glow-491104-q5.sales.sales202505`
UNION ALL SELECT * FROM `disco-glow-491104-q5.sales.sales202506`
UNION ALL SELECT * FROM `disco-glow-491104-q5.sales.sales202507`
UNION ALL SELECT * FROM `disco-glow-491104-q5.sales.sales202508`
UNION ALL SELECT * FROM `disco-glow-491104-q5.sales.sales202509`
UNION ALL SELECT * FROM `disco-glow-491104-q5.sales.sales202510`
UNION ALL SELECT * FROM `disco-glow-491104-q5.sales.sales202511`
UNION ALL SELECT * FROM `disco-glow-491104-q5.sales.sales202512`;
```

## 2. RFM CALCULATION (CTEs)

```sql
-- step2: Calculate recency, frequency, monetary, r, f, m ranks
-- combine views with CTEs

CREATE OR REPLACE VIEW `disco-glow-491104-q5.sales.rfm_metrics`
AS
WITH current_date AS (
  SELECT DATE('2026-03-25') AS analysis_date -- Today's Date
),
rfm AS (
  SELECT
    customerID,
    MAX(OrderDate) AS last_order_date,
    date_diff((SELECT analysis_date FROM current_date), MAX(OrderDate), DAY) AS recency,
    COUNT(*) AS frequency,
    SUM(OrderValue) AS monetary
  FROM `disco-glow-491104-q5.sales.sales2025`
  GROUP BY customerID
)
SELECT
  rfm.*,
  ROW_NUMBER() OVER(ORDER BY recency ASC) AS r_rank,
  ROW_NUMBER() OVER(ORDER BY frequency DESC) AS f_rank,
  ROW_NUMBER() OVER(ORDER BY monetary DESC) AS m_rank
FROM rfm;
```

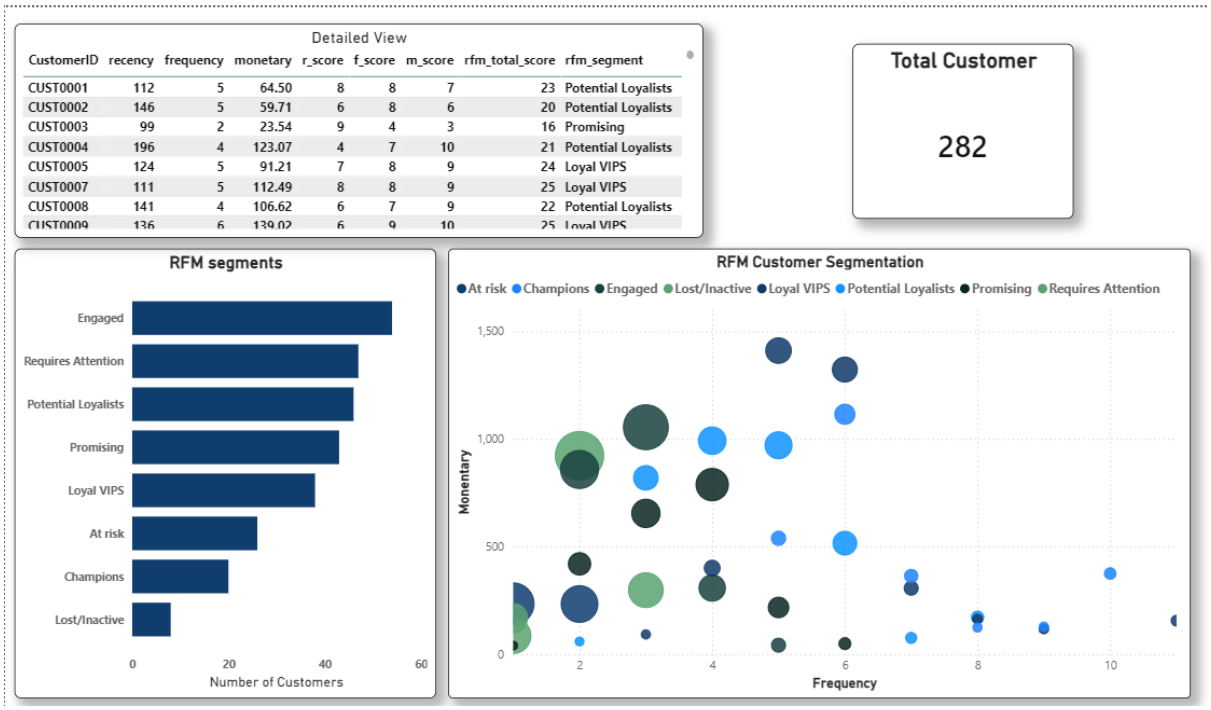
## 3. SCORING (NTILE Deciles)

```sql
-- step3: Assigning Deciles (10 = best, 1 = worst)
CREATE OR REPLACE VIEW `disco-glow-491104-q5.sales.rfm_scores`
AS
SELECT
  *,
  NTILE(10) OVER(ORDER BY r_rank DESC) AS r_score,
  NTILE(10) OVER(ORDER BY f_rank DESC) AS f_score,
  NTILE(10) OVER(ORDER BY m_rank DESC) AS m_score,
FROM `disco-glow-491104-q5.sales.rfm_metrics`;
```

## 4. BUSINESS SEGMENTS

```sql
-- step4 : Total Score
CREATE OR REPLACE VIEW `disco-glow-491104-q5.sales.total_score`
AS
SELECT
  CustomerID,
  recency,
  frequency,
  monetary,
  r_score,
  f_score,
  m_score,
  (r_score + f_score + m_score) AS rfm_total_score
FROM `disco-glow-491104-q5.sales.rfm_scores`
ORDER BY rfm_total_score DESC;
```

**5. POWER BI DASHBOARD**




| CustomerID | recency | frequency | monetary | r_score | f_score | m_score | rfm_total_score | rfm_segment |
|---|---|---|---|---|---|---|---|---|
| CUST0001 | 112 | 5 | 64.50 | 8 | 8 | 7 | 23 | Potential Loyalists |
| CUST0002 | 146 | 5 | 59.71 | 6 | 8 | 6 | 20 | Potential Loyalists |
| CUST0003 | 99 | 2 | 23.54 | 9 | 4 | 3 | 16 | Promising |
| CUST0004 | 196 | 4 | 123.07 | 4 | 7 | 10 | 21 | Potential Loyalists |
| CUST0005 | 124 | 5 | 91.21 | 7 | 8 | 9 | 24 | Loyal VIPS |
| CUST0007 | 111 | 5 | 112.49 | 8 | 8 | 9 | 25 | Loyal VIPS |
| CUST0008 | 141 | 4 | 106.62 | 6 | 7 | 9 | 22 | Potential Loyalists |
| CUST0009 | 136 | 6 | 139.02 | 6 | 9 | 10 | 25 | Loyal VIPS |

## BUSINESS INSIGHTS

• 81 customers (27%) drive 70%+ revenue

• Champions avg ₹170 vs overall ₹110

• At Risk customers dormant 300+ days

## RECOMMENDATIONS

1. Champions → VIP upsell (20 customers)

2. At Risk → 40% discount emails (23 customers)

3. Monthly auto-refresh pipeline

**EXPECTED ROI: +25% revenue from retention**

## TECHNICAL SKILLS DEMO

✅ BigQuery: CTEs/Window Functions/UNION ALL

✅ Power BI: DAX/Scatter Charts/Slicers/KPIs

✅ Analytics: RFM/CLV/Churn prediction

✅ End-to-end: Raw data → Actionable insights